

Docker Project Assignment - Python Web App

Scenario:

On your first working day as a DevOps engineer at EK_TECK SOFTWARE SOLUTION, you are assigned a project by your manager. The project has the following tasks and requirements:

Tasks / Requirements

- The application code is already written in Python and provided in the developer directory (s12student).
- Your base Docker image should be Ubuntu.
- Set the working directory inside the container to /app.
- Download and install python3 and python3-pip while writing your Dockerfile.
- Explain each step of your Dockerfile as you write it.
- The developer code includes:
 - /app/app.py - Python web server
 - /app/requirements.txt - Python dependencies
- Your container entrypoint will be python3, and your CMD should run app.py
- Deploy your application and access it in the browser at the /demo endpoint, for example:
http://<server-ip>:5000/demo/
- Push your Docker image to Docker Hub.